Human Activity Recognition from Thigh and Wrist Accelerometry

Alejandro Castellanos[1], **Antonio M. López**[2], **Diego García**[3], **Diego Álvarez**[4], **Juan C. Álvarez**[5]

Multisensor Systems and Robotics Lab (SiMuR). Electrical Engineering Department, University of Oviedo.

C/Pedro Puig Adam, Ed. Torres Quevedo (Departamental Oeste), bloque 2, 2.1.09 33204, Campus de Gijón, Spain.

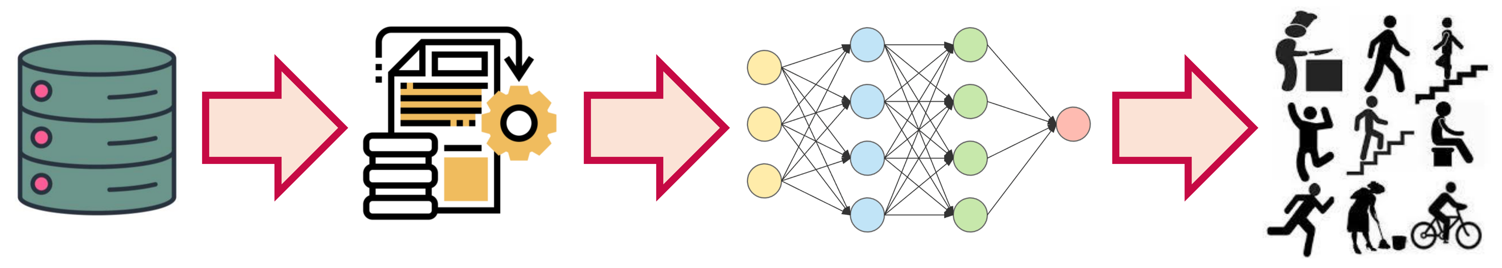
Contact details: [1]castellanosalejandro@uniovi.es, [2]amlopez@uniovi.es, [3]garciaperdiego@uniovi.es, [4]dalvarez@uniovi.es, [5]juan@uniovi.es

1. Introduction

In 2020, Spain's Instituto de Salud Carlos III (ISCIII) established the Infrastructure for Precision Medicine Associated with Science and Technology (IMPaCT), which will address the creation of a population-based cohort of approximately 200,000 individuals. Its objective will be to enable the integration of lifestyle data, clinical information and genetic information for the subsequent generation of predictive models to facilitate the effective implementation of Precision Medicine. The **IMPaCT cohort** includes assessments of physical activity, sedentary behaviour and physical fitness of participating individuals [1]. For 7 consecutive days, **biomechanical and physiological parameters** will be recorded using two **portable monitors** integrating a **triaxial accelerometer** and a **gyroscope** placed on both the wrist and thigh of the participants.

This study aims to shed light on the feasibility of identifying physical activity from **accelerometry at the thigh and wrist**, while analysing the effectiveness of simultaneous placement of the two monitoring devices. **Public datasets** have been used, covering a variety of activities. Subsequently, **Deep Convolutional Neural Network (CNN)** based structures have been applied for classification.

2. Methods



2.1.- Datasets

The datasets analysed in [2], selecting RSD: Realistic Sensor Displacement Dataset [3] and DSAD: Daily and Sport Activities Dataset [4] have been considered.

2.2.- Data Preprocessing

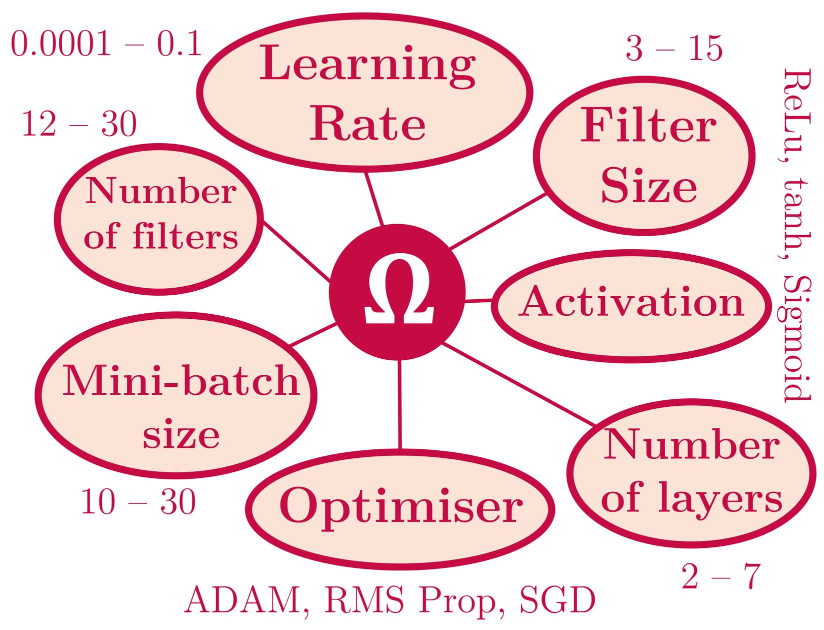
- ❑ Alignment with anatomical axes.
- ❑ RSD: 51585 data windows.
- ❑ DSA: 45443 data windows.
- ❑ Output of each window: mode.

Table 1.- Segmentation of datasets.

Resampling frequency	25 [Hz]
Segmentation windows	2 [s]
Window overlaps	1 [s]

2.3.- CNNs

- ❑ **"NET_TW"**: accelerations ($\vec{a}$) + rotational speeds ($\vec{\omega}$) on thigh and wrist.
- ❑ **"NET_T"**: $\vec{a} + \vec{\omega}$ on thigh.
- ❑ **"NET_W"**: $\vec{a} + \vec{\omega}$ on wrist.
- ❑ **Search space (Ω) of optimal hyperparameters.**



3. Results

After completing the search process, the 10 best combinations of **hyperparameters** were analysed. In all cases there is a similar score ($m \pm s = 0.93250 \pm 0.00066$). Therefore, we have selected those hyperparameters that simplify the model architecture; **minimising the number of hidden layers, the number of convolutional filters and their size.**

Table 2.- Optimal hyperparameters* of the classification model.

Number of Layers	Optimiser	Activation function	Mini-batch size	Number of filters	Filter Size	Learning Rate
2	RMS Prop	ReLu	10	12	7	0.0004549

* **Computation time** to optimise the hyperparameters of the model using **Random Search** [5]: **1 day, 12 hours, 14 minutes and 07 seconds.**

Table 3.- Accuracy (per cent) of Convolutional Neural Networks.

Accuracy	Train Data*	Validation Data	Test Data
NET_TW	0.8734	0.8685	**0.8524**
NET_T	0.7189	0.7141	0.6868
NET_W	0.7319	0.7316	0.6788

Optimal CNNs have ***7574 trainable parameters.***

* **Early Stopping** implemented in training. Patience = 5 epochs.

4. Conclusions and Discussion

- ✓ Classification based on **single limb** data ⟶ comparable accuracy (~ **68–69 [%]**).
- ✓ Accuracy is increased by 15 [%] when **combining thigh and wrist data.**
- ✓ **Non-atomic activities**: these are classified into various target classes (Table 5).
- ✓ Confusion during **transitions between activities.**

Table 4.- Extract of the confusion matrix (NET_TW, test data) for the target class 'Walking' (column).

	Walking
No Activity	283
Walking	1607
Walking (treadmill-flat)	429
Walking (treadmill-inclined)	49
Knees (alternating) to the breast	10
Going down (stairs)	118
Moving around in an elevator	49
Basketball	19

Future Work

Clustering
Establishment of 'Superclasses'.

Table 5.- Same as Table 4 for the class 'Basketball' (row).

	No Activity	Walking	Walking (treadmill-flat)	Walking (treadmill-inclined)	Running	Going up (stairs)	Going down (stairs)	Moving around in an elevator	Exercising on a stepper	Exercising on a cross trainer	Jumping	Basketball
Basketball	189	19	9	11	31	22	23	65	31	30	36	978

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